# Group 15 Hybrid RAG over Wikipedia

| | | F1 | Semantic Similarity | RR | NDCG@3 | Precision@3 | Type | response_time |
|---|---|---|---|---|---|---|---|---|
| 0 | which gene | 1 | 1 | 1 | 1 | 1 | factual | 10.9317 |
| 1 | s the linear | 1 | 1 | 1 | 0.9725 | 1 | inferential | 9.9784 |
| 2 | ;, for the sp | 1 | 1 | 1 | 0.9778 | 1 | factual | 10.8028 |
| 3 | the correla | 0.0656 | 0.5297 | 1 | 1 | 1 | comparative | 9.7164 |
| 4 | ay a very in | 1 | 1 | 0.5 | 0.9152 | 1 | comparative | 13.181 |
| 5 | rs in larger, | 0.7222 | 0.961 | 1 | 1 | 1 | factual | 10.1926 |
| 6 | adicals duri | 0.5625 | 0.789 | 1 | 1 | 1 | multi-hop | 12.8891 |
| 7 | ng isolation | 0.6111 | 0.672 | 1 | 0.9778 | 1 | comparative | 11.1737 |
| 8 | nd other fo | 0.0714 | 0.1411 | 0.5 | 0.9725 | 1 | comparative | 9.6196 |
| 9 | ries, allowir | 0.0606 | 0.5769 | 1 | 1 | 1 | factual | 10.2383 |

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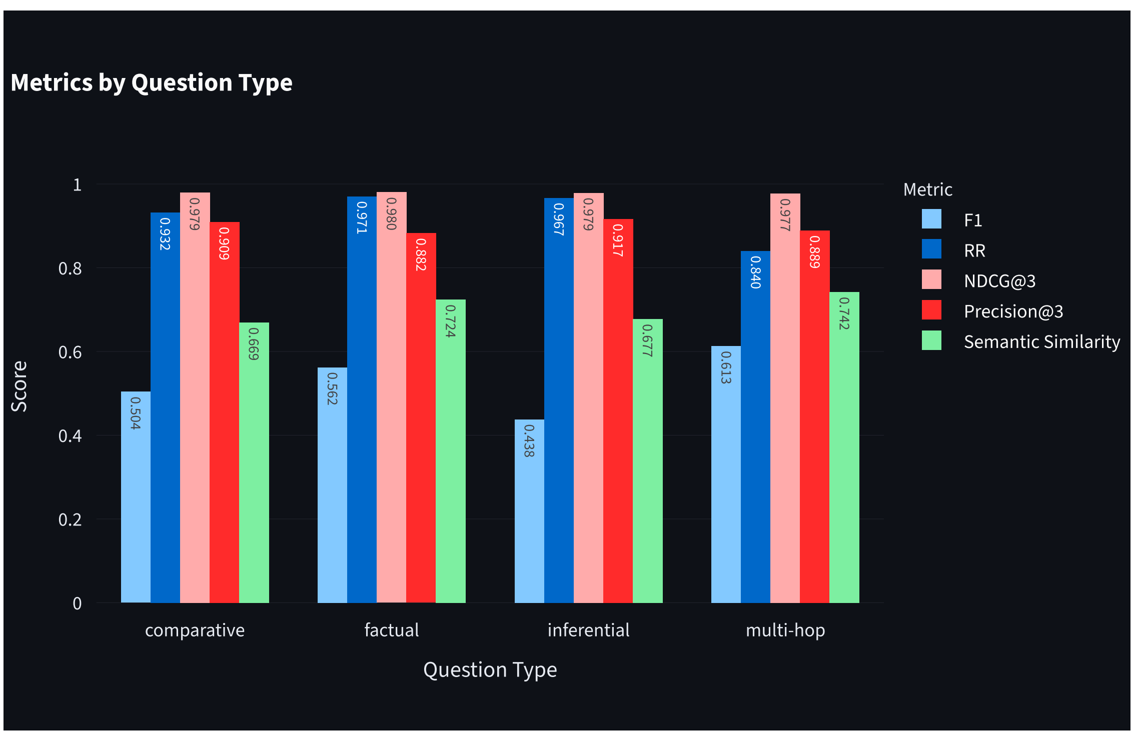


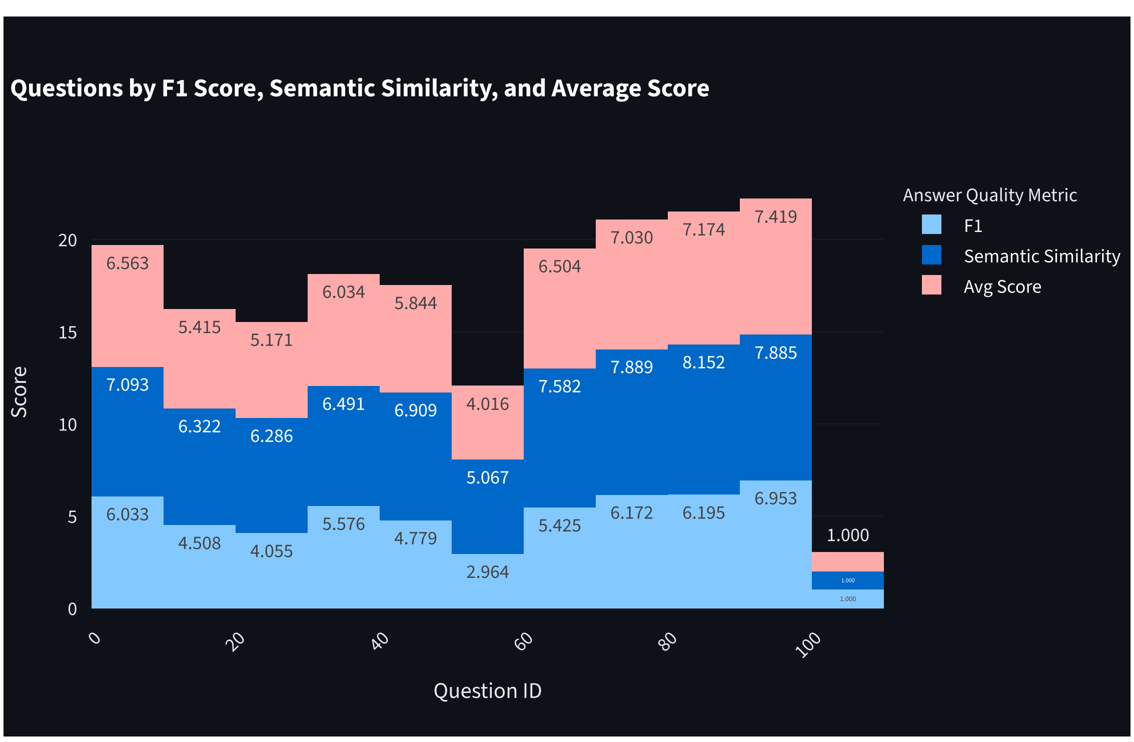

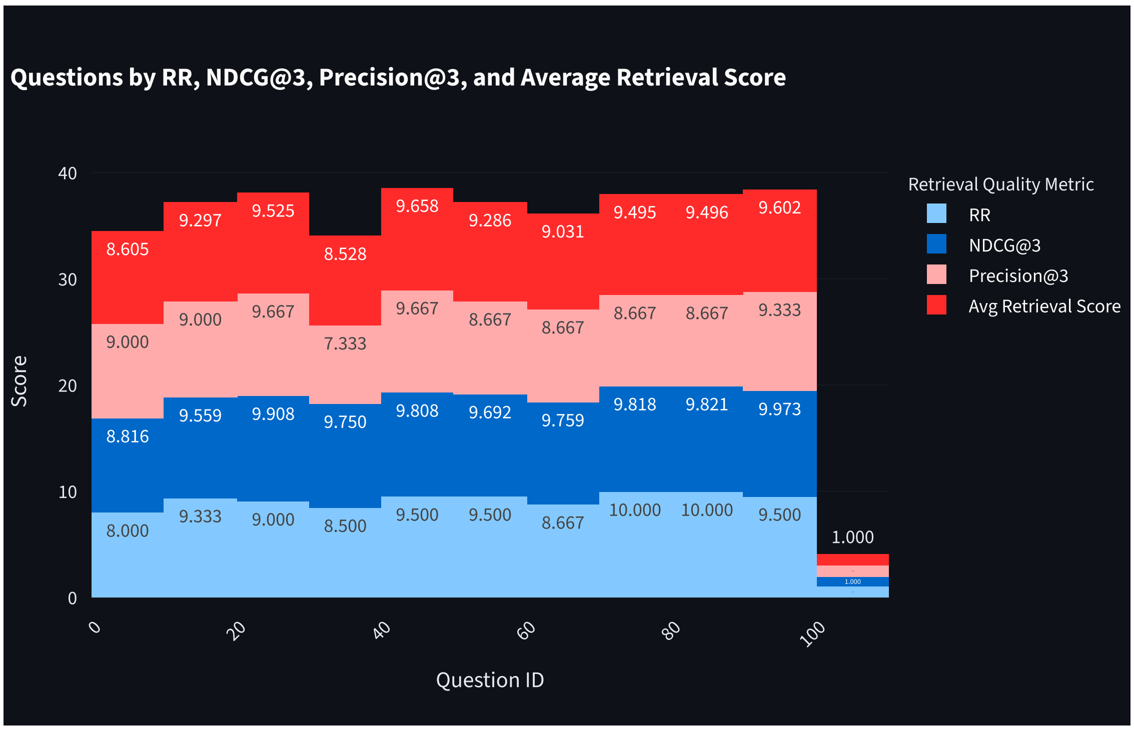


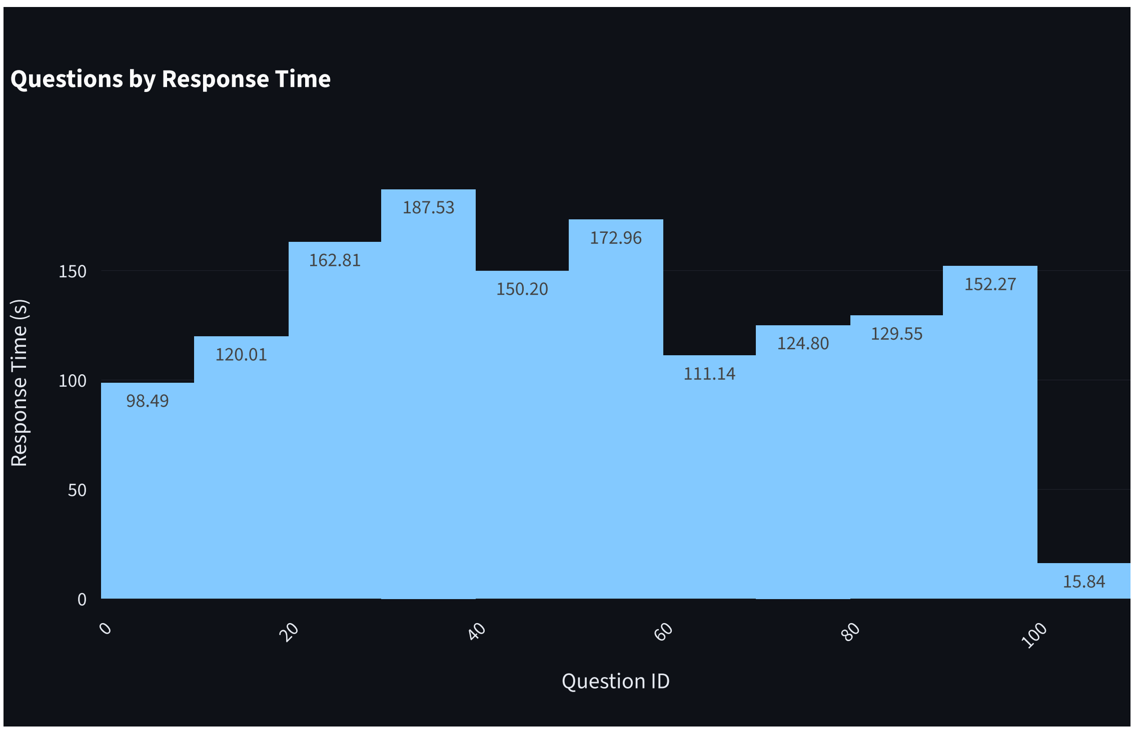


Enter your question

What is Geography ?

Top-K per retriever

3

Top-N after RRF

3

Ask

Mean Reciprocal Rank (URL): 0.7083 over 4 questions

Mean NDCG: 0.9124 over 4 questions at top 3 retrieved chunks

Mean Precision@3: 0.7500 over 4 questions

Mean Response Time: 9.31s over 4 questions

View information source: https://en.wikipedia.org/?curid=18963910

## Answer

The study of earth's seasons, climate, atmosphere, soil, streams, landforms, and oceans is a discipline that focuses on geography as an Earth science.

**Best supporting chunk ID:** 18963910_chunk_13
**Overlap:** 22
**Best URL contributing to answer:** https://en.wikipedia.org/?curid=18963910
**Reciprocal Rank:** 0.5000

Response time: 10.20s

# Top retrieved chunks (RRF)

**Chunk 1**

Geography has never been the exclusive preserve of geographers and has always posed a challenge to modern disciplinary thinking. As of April This geography-related article is a stub. You can help Wikipedia by adding missing information.


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**Chunk 2**

The interdisciplinary nature of the geographical approach depends on an attentiveness to the relationship between physical and human phenomena and their spatial patterns. While narrowing down geography to a few key concepts is extremely challenging, and subject to tremendous debate within the discipline, several sources have approached the topic. The 1st edition of the book "Key Concepts in Geography" broke down this into chapters focusing on "Space," "Place," "Time," "Scale," and "Landscape." The 2nd edition of the book expanded on these key concepts by adding "Environmental systems," "Social Systems," "Nature," "Globalization," "Development," and "Risk," demonstrating how challenging narrowing the field can be. Another approach used extensively in teaching geography are the Five themes of geography established by "Guidelines for Geographic Education: Elementary and Secondary Schools," published jointly by the National Council for Geographic Education and the Association of American Geographers in Just as all phenomena exist in time and thus have a history, they also exist in space and have a geography. — United States National Research Council, For something to exist in the realm of geography, it must be able to be described spatially. Thus, space is the most fundamental concept at the foundation of geography. The concept is so basic, that geographers often have difficulty defining exactly what it is. Absolute space is the exact site, or spatial coordinates, of objects, persons, places, or phenomena under investigation.


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**Chunk 3**

It is the newest of the branches, and often other terms are used in the literature to describe the emerging category. While human and physical geographers use the techniques employed by technical geographers, technical geography is more concerned with the fundamental spatial concepts and technologies than the nature of the data. It is therefore closely associated with the spatial tradition of geography while being applied to the other two major branches. These branches use similar geographic philosophies, concepts, and tools and often overlap significantly, so geographers rarely focus on just one of these topics, often using one as their primary focus and then incorporating data and methods from the other branches. Often, geographers are asked to describe what they do by individuals outside the discipline and are likely to identify closely with a specific branch, or sub-branch when describing themselves to lay people. Physical geography (or physiography) focuses on geography as an Earth science. It aims to understand the physical problems and the issues of lithosphere, hydrosphere, atmosphere, pedosphere, and global flora and fauna patterns (biosphere). Physical geography is the study of earth's seasons, climate, atmosphere, soil, streams, landforms, and oceans.


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